

Spring 2024

Assignment No. 2

BT503 - Environmet Biotechnology

By M.Kashif(03064963933)

Due Date: 04-07-2024

Assignment total marks: 10

Assignment Question:

Answer the questions after reading the review paper

Article title “Transformation of Waste Stabilization Ponds: Reengineering of an Obsolete Sewage Treatment System”

Access the article by clicking the link: <https://www.mdpi.com/2073-4441/13/9/1193>

Question#1: Briefly describe the importance of upflow anaerobic sludge blanket (UASB) reactor as mentioned in the Abstract and conclusion section of this published paper. **The word count limit is 50-100.**

Abstract

The Upflow Anaerobic Sludge Blanket (UASB) reactor offers several advantages over conventional Waste Stabilization Ponds (WSPs). The UASB reactor significantly reduces retention time and captures biogas, thereby mitigating greenhouse gas emissions and odor issues associated with WSPs. It produces high-quality effluent with low concentrations of biodegradable organic material, nutrients, and pathogens without requiring additional energy or materials.

Conclusion

The combination of a UASB reactor with sequential batch polishing ponds presents numerous benefits. This system greatly reduces the retention time and the area required for treatment compared to conventional WSPs. It efficiently removes organic material, pathogens, and nutrients, resulting in high-quality effluent. The absence of odor problems allows these systems to be situated closer to urban areas, thus reducing construction costs. Moreover, the sequential batch regime accelerates pathogen decay and nutrient removal, making the system more effective and space-efficient.

5 marks

Question#2: Write the followings **(5 marks)**:

- i. Journal name
- ii. Volume, Issue, and page numbers
- iii. Author names and their affiliations
- iv. Published date of the paper

Here are the corrected details based on the Article.

1) Journal Name: Water

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2) **Volume, Issue, and Page Numbers:** Volume 13, Issue 9, Page 1193

3) **Author Names and Their Affiliations:**

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4) **Published Date of the Paper:** April 25, 2021

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